Interfacing a Microprocessor to a Keyboard

A **keyboard** is an input device used to send data to a **microprocessor or computer**. When you press a key on the keyboard, you are actually **activating a switch**. The switch sends an electrical signal to the processor which tells the computer **which key was pressed**.

Different technologies are used to build these key switches. Each technology works in a slightly different way, but the main purpose is the same: **detect when a key is pressed and send that information to the processor**.

Types of Keyboard Switches

1. Mechanical key switches
2. Membrane key switches
3. Capacitive key switches
4. Hall effect key switches

1. Mechanical Key Switches

Mechanical switches are the **traditional type of keyboard switch**. In this type, pressing a key **physically connects two metal contacts**, which completes an electrical circuit and sends a signal to the processor.

Inside the switch, there is usually a **spring** that pushes the key back to its original position after you release it. Sometimes a small foam piece is also used to reduce the vibration or noise caused by the key pressing. In some modern designs, a **silicone rubber dome** with conductive material is used instead of metal contacts.

Working Principle

- Two metal contacts are inside the key switch.
- When the key is pressed, the contacts **touch each other**.
- This completes the circuit and sends a **key press signal**.
- When the key is released, the **spring returns the key to its normal position**.

Advantages

- Simple design
- Relatively **low cost**
- Easy to manufacture

Disadvantages

- **Contact bounce** occurs (the contact may open and close several times before settling).
- Contacts can become **dirty or oxidized with time**.
- Less reliable compared to some modern technologies.

The typical lifetime of mechanical switches is about **1 million keystrokes**, while silicone dome types may last about **25 million keystrokes**.

2. Membrane Key Switches

Membrane keyboards are a **special type of mechanical switch**. Instead of separate switches for each key, they use **three thin layers of plastic or rubber**.

The **top layer contains conductive lines for rows**, and the **bottom layer contains conductive lines for columns**. When a key is pressed, the top layer touches the bottom layer through a hole in the middle layer, completing the circuit.

Membrane keyboards are very common in **devices like calculators, microwaves, and cash registers** because they can be made **thin and sealed**.

Working Principle

- The keyboard has **three layers**.
- The **top layer has row conductors**.
- The **bottom layer has column conductors**.
- Pressing a key connects a **row line to a column line**.
- This connection sends a signal indicating a **key press**.

Advantages

- **Very thin design**
- Can be **sealed to protect from dust and water**
- Lower manufacturing cost

Disadvantages

- Shorter lifetime compared to some other switch types
- Less tactile feedback when typing

3. Capacitive Key Switches

Capacitive keyboards work using the **principle of capacitance** instead of physical electrical contacts.

Inside the switch, there are **two fixed metal plates** on the circuit board and another metal plate attached to a movable foam pad. When the key is pressed, the movable plate comes closer to the fixed plates, which **changes the capacitance**.

Special electronic circuits detect this change and produce a **logic signal** indicating that a key has been pressed.

Working Principle

- Two **fixed metal plates** are on the circuit board.
- A **movable metal plate** is attached to the key.
- Pressing the key **changes the capacitance** between the plates.
- A **sense amplifier detects the change** and generates a signal.

Advantages

- **No mechanical contact**, so no oxidation or dirt problems
- More **reliable and durable**

Disadvantages

- Requires **additional electronic circuitry** to detect capacitance changes.

These switches usually have a lifetime of about **20 million keystrokes**.

4. Hall Effect Key Switches

Hall effect switches work using a **magnetic field** instead of mechanical contacts. Because of this, they are very reliable.

A **semiconductor crystal** carries a small electric current. When the key is pressed, the crystal moves into a **magnetic field**. This magnetic field changes the movement of the electric charges in the crystal and produces a **small voltage** called the **Hall voltage**.

This voltage is amplified and used as the signal that a key has been pressed.

Working Principle

- A **current flows through a semiconductor crystal**.
- When the key is pressed, the crystal moves into a **magnetic field**.
- The magnetic field produces a **Hall voltage**.
- The voltage is **amplified and detected as a key press**.

Advantages

- **No mechanical contacts**
- Very **reliable**
- Extremely **long lifetime**

Disadvantages

- **More expensive**
- Requires **complex circuitry**

Hall effect keyboards can last about **100 million or more keystrokes**.

Keyboard Matrix Connection

In most keyboards, the switches are arranged in a **matrix of rows and columns**. Instead of connecting each key separately, keys share row and column lines.

When a key is pressed, it connects **one row line with one column line**. By detecting which row and column are connected, the system can determine **which key was pressed**.

To correctly read keyboard data, the system must perform three important tasks.

Major Tasks in Keyboard Interfacing

1. **Detect a key press**
2. **Debounce the key press**
3. **Encode the key press**

Software Keyboard Interfacing

In software interfacing, the **microprocessor program itself handles keyboard detection**.

The rows of the keyboard matrix are connected to **output port lines**, while the columns are connected to **input port lines**.

Normally, the column lines stay **HIGH because of pull-up resistors**. When a key is pressed, a row connects to a column and the signal becomes **LOW**, which the processor detects.

The processor repeatedly checks the rows and columns to see if any key has been pressed.

Steps of Software Keyboard Interfacing

1. Output **0 to all rows**.
2. Check if any **column becomes LOW**.
3. If a column is LOW, a **key press is detected**.

4. Wait about **20 ms for debounce**.
5. Check again to confirm the key press.
6. Determine the **row and column number**.
7. Convert the row-column value into a **code (like ASCII)**.

This method is called **two-key lockout**, meaning the system waits for the previous key to be released before detecting another key.

Hardware Keyboard Interfacing

Sometimes the CPU is very busy and cannot spend time scanning the keyboard. In such cases, a **special hardware device** is used to handle keyboard tasks.

One example is the **AY5-2376 keyboard encoder**.

This device automatically:

- Detects the key press
- Removes bounce
- Converts the key into an **ASCII code**

The encoded data is then sent to the microprocessor through **8 parallel lines**. When the data is ready, the encoder sends a **strobe signal** to the processor.

The processor can read the data either by:

- **Polling**, or
- **Interrupt method**

In the interrupt method, the CPU does not check the keyboard continuously. It only responds when the encoder sends an **interrupt signal**, which saves processor time.

Another advantage of this hardware is **two-key rollover**, meaning if two keys are pressed almost at the same time, both keys will still be detected correctly.

Exam Question

Interface a 4×4 keyboard with the Intel 8086 using the Intel 8255 Programmable Peripheral Interface and write an ALP to detect a key closure and return the key code in register AL. The debounce period for a key is 10 ms and a delay routine `DEBOUNCE` is available.

A **4×4 keyboard** has **16 keys** arranged in **4 rows and 4 columns**. Each key connects one **row line** with one **column line**.

When a key is pressed:

- The corresponding **row and column get electrically connected**.

- By checking **which row and which column are active**, the processor can determine **which key was pressed**.

To connect this keyboard with the **8086**, we use the **8255 PPI** because the 8086 itself does not provide enough direct I/O lines.

In this interface:

- **Port A → Output → used to select rows**
- **Port B → Input → used to read columns**

The processor activates rows **one at a time**, then reads the column lines. If a key in that row is pressed, the corresponding column becomes **LOW**, and the processor detects it.

The 8255 internally uses **two address lines (A0 and A1)** to select its internal registers.

A1	A0	Selected Register
0	0	Port A
0	1	Port B
1	0	Port C
1	1	Control Register

Address	A1 A0	Register
8000H	0	Port A
8002H	1	Port B
8004H	10	Port C
8006H	11	Control Word Register

If the base address is **8000H**, then:The only rule is that A1 A0 combinations must remain correct.

So the addresses depend on the **address decoder circuit**, not the chip itself.

Control word **82H** configures the ports of the 8255.

First convert it to **binary**:82H = 10000010.Control word format in **I/O mode**

Bit	Function
D7	Mode selection
D6 D5	Mode of Port A
D4	Port A direction
D3	Port C upper
D2	Port B mode
D1	Port B direction
D0	Port C lower

Now place the bits:

```
D7 D6 D5 D4 D3 D2 D1 D0
1  0  0  0  0  0  1  0
```

Meaning of Each Bit

D7 = 1

- Selects **I/O Mode,**Not BSR mode
  Because we need ports to transfer keyboard data, not just set/reset bits.

D6 D5 = 00

- **Mode 0 for Port A**

Mode 0 is selected because: Keyboard scanning is simple. No handshaking required.We only read key states

D4 = 0

Port A = **Output.** Port A sends signals to **activate keyboard rows**.

D3 = 0

Port C upper = **Output.** (Not used but kept output to avoid floating inputs)

D2 = 0

Port B Mode = **Mode 0**

D1 = 1

Port B = **Input,** Because Port B reads the **column signals** from the keyboard.

D0 = 0

Port C lower = **Output ,**(Not used in this design)

Port	Direction
Port A	Output
Port B	Input
Port C	Output

Mode = **Mode 0,** Thus control word = **82H**

Why Mode 0 for keyboard

Keyboard scanning only requires:

- sending row signals
- reading column signals

No synchronization signals are required.

Therefore **Mode 0 is sufficient and simplest**.**Mode 1 includes:**

- handshaking signals
- interrupt logic

But keyboard scanning **does not need handshaking**, so Mode 1 would be **unnecessary and more complex**.

What if Port A was Input?-Then the processor **cannot send row selection signals**, so keyboard scanning **will not work**.

What if Port B was Output?-Then the processor cannot read column signals, so key presses cannot be detected.

Thus the correct configuration is:Port A → Output,Port B → Input

Key Debouncing

Mechanical keys do not produce a clean signal immediately.

When pressed, the contacts **bounce** and produce multiple transitions.

This may cause the processor to detect **multiple key presses**.

To prevent this, a **10 ms delay** is inserted after detecting a key press.

The delay routine **DEBOUNCE** waits **10 milliseconds**, then the key is checked again to confirm the press.

ALP Logic (Concept)

Steps used by the program:

1. Initialize 8255 with control word **82H**
2. Send row pattern to **Port A**
3. Read **Port B**
4. If any column becomes **LOW**, key detected
5. Call **DEBOUNCE routine**
6. Check again
7. Identify row and column
8. Generate key code
9. Store key code in **AL**